

THEMBELIHLE MOLOPE

ICT Graduate & CAPACITI Demand 2 Digital Associate

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PROFESSIONAL PROFILE

I am an ICT graduate from Durban University of Technology, graduating with a 73.75% average, currently on the CAPACITI Demand 2 Programme where I am gaining hands-on experience in software development, data analytics and AI-driven technologies. I have built full-stack applications from scratch, developed AI-powered tools using Gemini and OpenAI APIs, and completed certifications in Generative AI, Prompt Engineering, and Responsible AI from IBM, Google, and DeepLearning.AI. I bring practical experience across Python, SQL, Business Intelligence, React, Node.js, and FastAPI, alongside full-stack development in C#, ASP.NET, MySQL, and JavaScript, and cloud fundamentals in Microsoft Azure. I am curious, fast-moving, and genuinely passionate about building technology that creates meaningful impact.

EXPERIENCE

CAPACITI Demand 2 Digital Associate | August 2026 – Present

CAPACITI — Braamfontein, Johannesburg

- Completing intensive full-stack development and data analytics training at the CAPACITI Braamfontein Accelerator.
- Built AI-powered applications using Gemini and OpenAI APIs, applying responsible AI principles throughout.
- Completed professional development workshops in communication, financial literacy, and workplace readiness.
- Collaborating in a team environment on real-world AI and software projects aligned to industry standards.

EDUCATION

Advanced Diploma in Information and Communication Technology

Durban University of Technology

- NQF Level 7 | Status: In Progress (Expected: 2026)
- Focus Areas: Software Development & Management, Business Intelligence, Data Structures, Platform Based Development, Applied Mathematics (Statistics & Discrete Structures), Strategy Acquisition & Management, Research Skills

National Diploma in Information and Communication Technology

Durban University of Technology

- NQF Level 6 | Majors: Application Development and Information Systems | 2025
- Overall Average: 73.75% | Distinction Rate: 65.5%
- Focus Areas: Applications Development, Information Systems, Mobile Computing, Cloud Computing (Azure), IT Project Management, Computer Security, Communication Networks, Business Fundamentals, Agile & SDLC

Higher Certificate in Information Technology (Cum Laude)

Durban University of Technology

- NQF Level 5 | 2022
- Focus Areas: IT Technical Support, Hardware & Software Support, Networking, Database Administration, E-Commerce, Solutions Development (Java), Web Development & Web Technologies

AI PROJECTS - CAPACITI

StudyBuddy AI | Node.js, Express, HTML, CSS, JavaScript, Gemini API, OpenAI API

- Built a beginner-friendly AI study assistant supporting topic explanations, summaries, study planning, quiz generation, and exam prep.
- Integrated Gemini and OpenAI APIs with server-side validation, responsible AI guardrails, and PDF, DOCX, and image upload support.
- Implemented downloadable study exports, theme switching, and persistent chat history as a full browser-based application.

LOPE-LEE | AI Content Studio | *React, FastAPI, Python, Tailwind CSS, OpenAI API*

- Designed and built a full-stack AI content generation studio for creating social posts, blog outlines, emails, and ad copy.
- Built a FastAPI Python backend with prompt engineering, content validation, and OpenAI-compatible API integration.
- Delivered a polished React frontend with dark/light themes, markdown rendering, copy-to-clipboard, and persistent generation history.

ACADEMIC PROJECTS

Reverse Vending Machine Waste Management System | *C#, ASP.NET MVC, MySQL, HTML, CSS, JavaScript*

- Developed a web-based recycling management system supporting DUT's Envision2030 sustainability initiative.
- Built backend logic in C# with ASP.NET MVC and integrated a MySQL database for data management.

EmpowerHer – Doctor Booking Appointment System | *JavaScript, C#, ASP.NET, HTML, CSS, MySQL*

- Designed a digital platform enabling women to schedule and manage medical appointments easily.
- Integrated a real-time database for booking updates and secure patient data management, with a strong focus on accessibility.

RoadToRedemption – Ex-Offenders Reintegration System | *C#, ASP.NET, MySQL, CSS*

- Built a digital solution supporting ex-offenders with job placement, digital skills development, and mentorship access.

E-Commerce Web Project | *HTML, MySQL, JavaScript, CSS*

- Developed an online shopping platform with secure user authentication, database-driven inventory control, and checkout functionality.

TECHNICAL SKILLS

Languages: Python, C#, Java, JavaScript, SQL, HTML, CSS

AI & Emerging Tech: Generative AI, Prompt Engineering, OpenAI API, Gemini API, Responsible AI

Frontend & Frameworks: React, Tailwind CSS, ASP.NET MVC, Node.js, Express, FastAPI

Data & Databases: MySQL, SQL, Azure Cosmos DB, Power BI, Business Intelligence, Statistics

Cloud & Platforms: Microsoft Azure, AWS Cloud Fundamentals

Concepts & Practices: SDLC, Agile, Technical Writing, Data Reporting & Analytics, Research Skills

Tools & Productivity: GitHub, Visual Studio, VS Code, Postman, Microsoft Office

CORE COMPETENCIES

- Fast Learner & Continuous Improvement
- Problem Solving & Critical Thinking
- Teamwork & Collaboration
- Time Management & Works Well Under Pressure
- Active Listening & Clear Communication
- Leadership, Initiative & Interpersonal Skills
- Research Skills
- Resilience & Adaptability in Fast-Paced Environments

CERTIFICATIONS

- Introduction to Cybersecurity — Cisco Networking Academy
- NDG Linux Unhatched — Cisco Networking Academy
- Full-Stack Development Certificate — FNB Academy

LEADERSHIP & COMMUNITY

Student Volunteer Trainer — ICT Skills Training for Inmates

Pietermaritzburg Medium A Correctional Centre | 2024

- Designed and delivered an ICT skills training programme for inmates to improve employability upon release.
- Mentored 26 inmates, improving digital literacy and confidence, while collaborating with prison staff to align training with rehabilitation goals.
- Demonstrated patience, empathy, and the ability to communicate effectively with people from all walks of life.

LANGUAGES

- English, IsiZulu, Sepedi

REFERENCES

Available on request.